

8 Scientists use many different techniques in genetic engineering.

- (a) Sometimes the gene for genetic engineering cannot be extracted from the donor organism. Instead, the gene is synthesised using one of two different methods.

Outline the **two** methods for synthesising a gene for use in genetic engineering.

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..... [3]

- (b) DNA ligase and DNA polymerase are two enzymes that are used in genetic engineering.

Complete Table 8.1 to show the roles of DNA ligase and DNA polymerase in genetic engineering.

Use a tick (✓) if the enzyme has the role or a cross (X) if the enzyme does **not** have the role.

Table 8.1

| role in genetic engineering                             | DNA ligase | DNA polymerase |
|---------------------------------------------------------|------------|----------------|
| joins two sections of sugar phosphate backbone in DNA   |            |                |
| adds a gene to a plasmid                                |            |                |
| adds free activated DNA nucleotides to a polynucleotide |            |                |

[3]

(c) The polymerase chain reaction (PCR) is used to make many copies of a gene.

Three temperatures are used in a PCR cycle.

State the three temperatures that are used, **and** outline what happens at each temperature during a PCR cycle.

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[Total: 9]